1. Write short answers to the following questions:

a. What is meant by "database"?
b. Why do organizations store data?
c. Define "data" in one line.
d. Define "information" in one line.
e. What does a DBMS do?
f. Write any two examples of DBMS.
g. State one goal of a DBMS regarding duplication.
h. What does the INT data type represent?
i. What is the primary use of VARCHAR?
j. What does the DATE data type store?
k. What is a table in a database?
l. What is a record (tuple)?
m. What is a field (column)?
n. What is a primary key?
o. What is a foreign key?
p. What is referential integrity?
q. What does the CREATE command do?
r. What does the ALTER command do?
s. What does the DROP command do?
t. What does the SELECT command do?
u. Write about AND Operator.
v. Why use OR operator?

2. Write the long answer to the following questions:

a. Explain briefly how databases aid decision-making.
b. How does a DBMS speed up searching?
c. Why are backups important in databases?
d. Differentiate between data and information with a simple example.
e. Why is choosing the correct data type necessary?
f. When would you prefer DECIMAL over FLOAT?
g. When is CHAR more suitable than VARCHAR?
h. What kind of content suits VARCHAR(MAX)?
i. Why should every table have a primary key?
j. Why do we relate tables using foreign keys?
k. How does a DBMS allow safe multi-user access?
l. What risk comes with dropping a table?
m. Why might we alter a table after deployment?
n. Give one practical reason to update existing data.
o. What precaution is needed before deleting rows?
p. Why are queries referred to as "questions to the database"?
q. How do reports relate to queries in real use?
r. Why reduce duplicate data in databases?
s. Why should changes be instantly visible to users?
t. Why does the accuracy of stored data matter?
u. Write the importance of BETWEEN operator.

1. Write short answers to the following questions

a. What is meant by "database"?

A database is an organized collection of related data stored electronically for easy access and management.

b. Why do organizations store data?

Organizations store data to keep records, make decisions, and manage their operations efficiently.

c. Define "data" in one line.

Data is a collection of raw facts and figures.

d. Define "information" in one line.

Information is processed and organized data that is meaningful.

e. What does a DBMS do?

A DBMS (Database Management System) stores, organizes, retrieves, updates, and manages data efficiently.

f. Write any two examples of DBMS.

The two examples of DBMS are:

- MySQL
- Oracle Database

g. State one goal of a DBMS regarding duplication.

A DBMS goal is to reduce data duplication (redundancy).

h. What does the INT data type represent?

The INT data type stores whole numbers (integers).

i. What is the primary use of VARCHAR?

VARCHAR is used to store variable-length text or characters.

j. What does the DATE data type store?

The DATE data type stores dates in the format YYYY-MM-DD.

k. What is a table in a database?

A table is a collection of related data arranged in rows and columns.

l. What is a record (tuple)?

A record (tuple) is a single row in a table containing information about one item.

m. What is a field (column)?

A field (column) is a single attribute or category of data in a table.

n. What is a primary key?

A primary key is a field that uniquely identifies each record in a table.

o. What is a foreign key?

A foreign key is a field that links one table to another by referring to the primary key of another table.

p. What is referential integrity?

Referential integrity ensures that relationships between tables remain valid and consistent.

q. What does the CREATE command do?

The CREATE command is used to create database objects such as databases and tables.

r. What does the ALTER command do?

The ALTER command is used to modify the structure of an existing table.

s. What does the DROP command do?

The DROP command permanently deletes a database object such as a table or database.

t. What does the SELECT command do?

The SELECT command retrieves data from one or more tables.

u. Write about AND Operator.

The AND operator returns records only when all specified conditions are true.

v. Why use OR operator?

The OR operator returns records when at least one of the specified conditions is true.

2. In a long answer, respond to the following questions:

a. Describe in short a role of a database in decision making.

Databases contain huge quantities of accurate and meaningful data which assist users to easily locate the information they need, create reports, and analyze data and helps organizations to make informed and timely decisions.

b. What happens to the speed of searching if a DBMS is used?

A DBMS is a program that organizes the data and provides the ability to query the data which enables us to locate specific information much quicker than manually searching.

c. Why is it important to make backups in databases?

The backup is useful because it safeguards the data from getting lost because of hardware failure, software mistakes, viruses or accidental deletion. In case data is lost, it can be recovered from the back up.

d. Differentiate between data and information with a simple example.

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DATA VS. INFORMATION

CHARACTERISTIC	DATA	INFORMATION
Definition	Raw, unorganized facts with no inherent meaning	Data that has been processed, cleaned, and given context to make it meaningful
Form	Contact names, email addresses, time entry logs, event RSVPs	Segmented client lists, engagement reports, relationship maps, dashboards
Context	None. No clear link between entries or relevance to firm priorities	Answers "who, what, when, where" and shows how it relates to firm goals or matters
Purpose	Record events, transactions, or interactions	Explain or describe patterns, identify key segments, reveal areas of focus
Example	5,000 unverified email addresses exported from attorneys' Outlook address books	500 verified contacts in the finance sector, segmented by seniority and relationship strength, ready for targeted outreach
Value to the Firm	Minimal on its own. It can't drive strategy without further processing	Actionable for BD, client retention, and event targeting

e. Why is it important to select the right type of data?
Selecting the right data type aids accurately storing data, save storage space, increase its performance, and lower errors during data entry.

f. When would you prefer DECIMAL over FLOAT?
When precise values are needed, as in financial or banking calculations, DECIMAL is preferred to FLOAT, since it is more precise.

g. When to use CHAR vs VARCHAR?
CHAR is better for use when the data does not vary in length, like a gender code (M/F) or a country code, due to the fixed length storage.

h. Which type of data is suitable for VARCHAR(MAX)?
VARCHAR(MAX) is for storing a very large text, e.g. descriptions, comments, articles, notes, etc. which may be larger than the usual text size.

i. Why is it that all tables must have a primary key?
Each table should have a primary key since it will uniquely identify each record in the table. It eliminates duplicate records and simplifies data searching and updating.

j. Why use foreign keys to relate tables?
Relationships are made between tables with foreign keys. They ensure consistency of data and ensure that related data is accurate.

k. How can a DBMS ensure multiple users can access the system safely?

Several users can access the database simultaneously and can control access, and prevent conflicts, using a DBMS. This ensures the data is safe and reliable.

l. What is the danger of dropping a table?
When a table is dropped, all the data in the table will be deleted as well as the table itself. After deletion, the information can only be retrieved if a backup exists.

m. What could be reasons for changing a table after deployment?
A table can be modified even after deployment by adding new columns and removing columns that are not required or changing the data type of existing columns to meet changing requirements.

n. Provide one real-world justification for updating of any data.
Current data is updated to maintain accurate and up-to-date records. For instance, if there are changes in the customer's address or phone number, this may need to be updated.

o. Which is the right precaution to take before removing rows?
When deleting rows, it is important to check that the data that has been selected is the correct data and also make sure to take a back up copy of the data to prevent unintentional loss.

p. Why is it that a query is also called a "question to the database"?
Queries are referred to as "questions to the database" because they ask for specific information from the database, which then returns the relevant information.

q. In practice, how do reports relate to queries?
User queries fetch the information from the database and reports display that information in a user-friendly manner.

r. So why store redundant data in databases?
Eliminating duplicate data frees up storage space, enhances data quality and simplifies database administration.

s. Why does it make sense to users to have to wait to see changes?
Changes should be immediately visible to ensure that everyone is using the most up to date and accurate information, eliminating confusion and wrongdoing.

t. Why is it important that the data in storage is accurate?
Because Data is essential for creating reports, making decisions and avoiding errors.

u. Write the importance of BETWEEN operator.
To get values within a range, use the BETWEEN operator. It simplifies queries, makes them more easily understood, and lessens multiple conditions.

These answers are suitable for examination questions that require 5–10 marks: they are well presented, clear and informative, but not too wordy.

4. Full Forms

1. **DB** – Database
2. **DBMS** – Database Management System
3. **RDBMS** – Relational Database Management System
4. **DBA** – Database Administrator
5. **PK** – Primary Key
6. **FK** – Foreign Key
7. **SQL** – Structured Query Language
8. **DDL** – Data Definition Language
9. **DML** – Data Manipulation Language
10. **TCL** – Transaction Control Language
11. **DCL** – Data Control Language

5. Multiple Choice Questions

1. **b. A structured collection of related data**
2. **c. Authorized access**
3. **b. One-to-Many**
4. **c. Junction table**
5. **c. Information**
6. **c. Replace raw data with processed information automatically**
7. **b. INT**
8. **b. INSERT**
9. **c. SELECT**
10. **a. DROP TABLE**
11. **c. Uniquely identifies each record in a table**
12. **b. Links a field in one table to the primary key of another**
13. **b. UPDATE**
14. **c. DELETE**
15. **c. A student has one library card**
16. **c. Tuple/Record**
17. **b. ALTER**
18. **a. CREATE DATABASE**
19. **c. VARCHAR**
20. **b. Prevents a column from containing NULL values**
21. **c. Social media browsing**
22. **a. DROP DATABASE**
23. **b. Data**
24. **a. UPDATE**
25. **a. Many students belong to one class**

26. **c. Database accuracy**
27. **b. Foreign key**
28. **b. Rows and columns**
29. **b. Ensures data is stored securely and retrievable efficiently**
30. **a. SELECT * FROM table_name;**

3. SQL Commands

a. List all columns and values for every student.
SELECT * FROM Student;

b. Show student name and grade of students whose grade is greater than 80.
SELECT student_name, grade
FROM Student
WHERE grade > 80;

c. Identify students whose email value is NULL.
SELECT *
FROM Student
WHERE email IS NULL;

d. Arrange all departments in ascending order of department name.
SELECT *
FROM Department
ORDER BY dept_name ASC;

e. Count how many students belong to each department.
SELECT dept_id, COUNT(*) AS student_count
FROM Student
GROUP BY dept_id;

f. List course code and title for courses offered by the Computer Science department.
SELECT c.course_code, c.title
FROM Course c
JOIN Department d ON c.dept_id = d.dept_id
WHERE d.dept_name = 'Computer Science';

g. Find the names of students who are enrolled in the course titled DBMS.
SELECT s.student_name
FROM Student s
JOIN Enrollments e ON s.student_id = e.student_id
JOIN Course c ON e.course_id = c.course_id
WHERE c.title = 'DBMS';

h. For each course, determine the number of enrolled students.

```
SELECT c.course_id, c.title, COUNT(e.student_id) AS enrolled_students
FROM Course c
LEFT JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_id, c.title;
```

i. Identify students who are not enrolled in any course.

```
SELECT s.*
FROM Student s
LEFT JOIN Enrollments e ON s.student_id = e.student_id
WHERE e.student_id IS NULL;
```

j. Update the grade of student_id = 5 to 78.50.

```
UPDATE Student
SET grade = 78.50
WHERE student_id = 5;
```

k. Add a new student with the given details.

```
INSERT INTO Student
(student_id, student_name, grade, email, dept_id)
VALUES
(7, 'Nirmala', NULL, 'nirmala@example.com', 30);
```

l. Remove all enrollments in semester 2025SP for course CS101.

```
DELETE FROM Enrollments
WHERE course_id = (
   SELECT course_id
   FROM Course
   WHERE course_code = 'CS101'
)
AND semester = '2025SP';
```

m. List each student with their department name and grade, sorted from highest grade to lowest.

```
SELECT s.student_name, d.dept_name, s.grade
FROM Student s
JOIN Department d ON s.dept_id = d.dept_id
ORDER BY s.grade DESC;
```

n. Compute the average grade for each department, ignoring NULL grades.

```
SELECT dept_id, AVG(grade) AS average_grade
FROM Student
GROUP BY dept_id;
```

o. List students whose name starts with the letter 'C'.

```
SELECT *
FROM Student
WHERE student_name LIKE 'C%';
```

p. Make the email field mandatory for every student.

```
ALTER TABLE Student
MODIFY email VARCHAR(100) NOT NULL;
```

q. Rename the column student_name to sname.

```
ALTER TABLE Student
RENAME COLUMN student_name TO sname;
```

r. Ensure that course_code is unique across all courses.

```
ALTER TABLE Course
ADD CONSTRAINT uq_course_code UNIQUE (course_code);
```

s. Remove the uniqueness constraint on course_code.

```
ALTER TABLE Course
DROP INDEX uq_course_code;
```

t. For every student, show how many courses they have taken, including students with zero courses.

```
SELECT s.student_id, s.student_name,
       COUNT(e.course_id) AS course_count
FROM Student s
LEFT JOIN Enrollments e
ON s.student_id = e.student_id
GROUP BY s.student_id, s.student_name;
```

To practice the upper commands run the following codes first.

```sql
CREATE DATABASE SchoolDB;

USE SchoolDB;

-- Department Table
CREATE TABLE Department (
   dept_id INT PRIMARY KEY,
   dept_name VARCHAR(50) NOT NULL
);

-- Department Data
INSERT INTO Department (dept_id, dept_name)
VALUES
(10, 'Computer Science'),
(20, 'Management'),
(30, 'Economics');


-- Student Table
CREATE TABLE Student (
   student_id INT PRIMARY KEY,
   student_name VARCHAR(50) NOT NULL,
   grade DECIMAL(5,2),
   email VARCHAR(100),
   dept_id INT,
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

-- Student Data
INSERT INTO Student
(student_id, student_name, grade, email, dept_id)
VALUES
(1, 'Ram', 85, 'ram@example.com', 10),
(2, 'Sita', 92.5, 'sita@example.com', 10),
(3, 'Bimal', NULL, 'bimal@example.com', 20),
(4, 'Chitra', 88, NULL, 20),
(5, 'Deepak', 75, 'deepak@example.com', 30),
(6, 'Laxmi', NULL, 'laxmi@example.com', 20);


-- Course Table
CREATE TABLE Course (
   course_id INT PRIMARY KEY,
```

```sql
    course_code VARCHAR(20),
    title VARCHAR(100),
    dept_id INT,
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

-- Course Data
INSERT INTO Course
(course_id, course_code, title, dept_id)
VALUES
(100, 'CS101', 'Intro to CS', 10),
(101, 'CS102', 'DBMS', 10),
(200, 'MG201', 'Principles Mgmt', 20),
(300, 'EC210', 'Microeconomics', 30);


-- Enrollments Table
CREATE TABLE Enrollments (
    student_id INT,
    course_id INT,
    semester VARCHAR(20),
    PRIMARY KEY (student_id, course_id, semester),
    FOREIGN KEY (student_id) REFERENCES Student(student_id),
    FOREIGN KEY (course_id) REFERENCES Course(course_id)
);

-- Enrollment Data
INSERT INTO Enrollments
(student_id, course_id, semester)
VALUES
(1, 100, '2025SP'),
(1, 101, '2025SP'),
(2, 101, '2025SP'),
(3, 200, '2025SP'),
(5, 100, '2025SP'),
(6, 200, '2025SP');
```